

HDOC DAY – 17

Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer. Define one separate user defined function for each menu option, with void return type and with a simple function call (without call by value/reference). Each function should read the required input, process it, and display the result.

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

The program should perform the following operations:

1. Generates the largest possible number using its digits.
2. Generates the smallest possible number using its digits.
3. Swap its first and last digits.
4. Checks whether it is a palindrome number.

AIM:

To write a menu-driven C program using switch-case and separate user defined function with void return type to perform digit-based operations on a positive integer, such as finding the largest and smallest possible number using its digits, swapping the first and last digits, and checking whether the number is a palindrome.

ALGORITHM

1. Start
2. Display the menu:
 1. Largest possible number
 2. Smallest possible number
 3. Swap first and last digits
 4. Check palindrome
3. Read the user choice.
4. Use switch-case according to the choice.
5. Call the corresponding user-defined function.
6. The selected function reads the positive integer.
7. Perform the required digit operation.
8. Display the result.
9. Stop

Test Case Menu Choice Input Number Expected Output

1	1	4213	Largest number = 4321
2	2	4213	Smallest number = 1234
3	3	12345	After swapping = 52341
4	4	121	Palindrome number
5	4	123	Not a palindrome number

>Write a code C.

```
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day17.c Untitled2 x
1 #include <stdio.h>
2
3 void largest();
4 void smallest();
5 void swapdigits();
6 void palindrome();
7
8 int main()
9 {
10     int choice;
11
12     printf("1. Largest possible number\n");
13     printf("2. Smallest possible number\n");
14     printf("3. Swap first and last digits\n");
15     printf("4. Check palindrome\n");
16
17     printf("Enter your choice: ");
18     scanf("%d", &choice);
19
20     switch(choice)
21     {
22         case 1:
23             largest();
24             break;
25
26         case 2:
27             smallest();
28             break;
29
30         case 3:
31             swapdigits();
32             break;
33
34         case 4:
35             palindrome();
36             break;
37
38         default:
39             printf("Invalid choice");
40     }
41
42     return 0;
43 }
44
45 void largest()
46 {
47     int n, a[10], i = 0, j, k, temp;
48
49     printf("Enter number: ");
50     scanf("%d", &n);
```

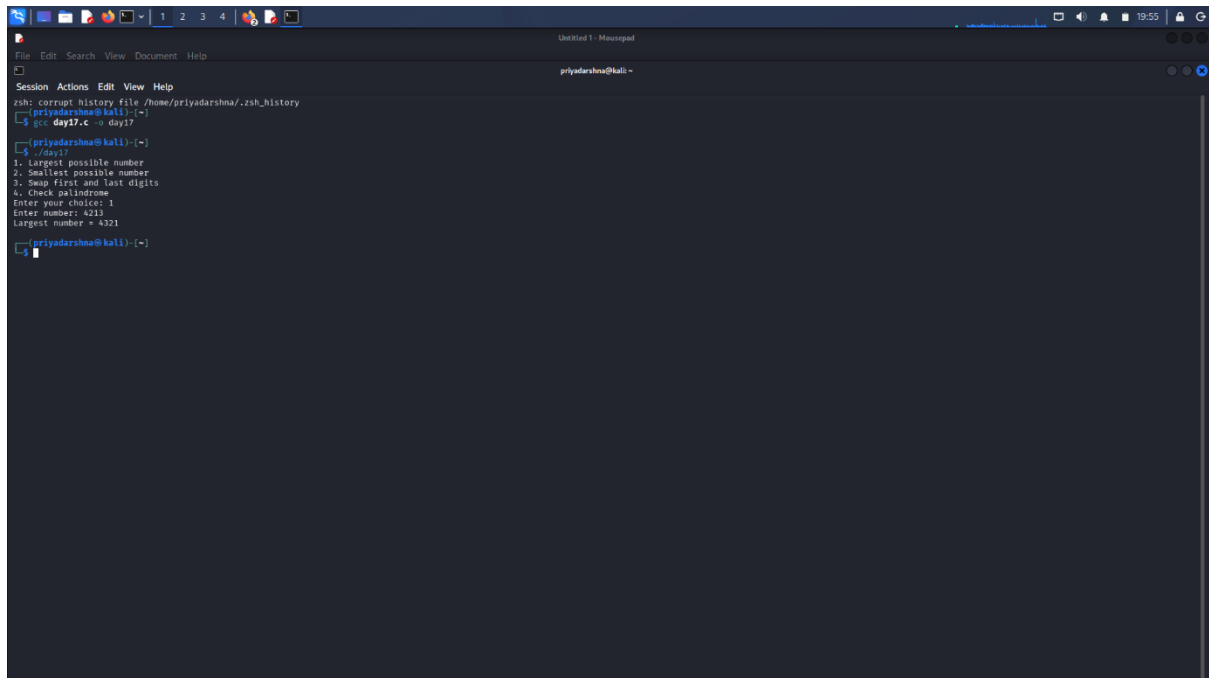
```
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day17.c Untitled2 x

51 while(n > 0)
52 {
53     a[i++] = n % 10;
54     n = n / 10;
55 }
56
57 for(j = 0; j < i; j++)
58     for(k = j + 1; k < i; k++)
59         if(a[j] < a[k])
60         {
61             temp = a[j];
62             a[j] = a[k];
63             a[k] = temp;
64         }
65
66 printf("Largest number = ");
67 for(j = 0; j < i; j++)
68     printf("%d", a[j]);
69 }
70
71 void smallest()
72 {
73     int n, a[10], i = 0, j, k, temp;
74
75     printf("Enter number: ");
76     scanf("%d", &n);
77
78     while(n > 0)
79     {
80         a[i++] = n % 10;
81         n = n / 10;
82     }
83
84     for(j = 0; j < i; j++)
85         for(k = j + 1; k < i; k++)
86             if(a[j] > a[k])
87             {
88                 temp = a[j];
89                 a[j] = a[k];
90                 a[k] = temp;
91             }
92
93     printf("Smallest number = ");
94     for(j = 0; j < i; j++)
95         printf("%d", a[j]);
96 }
97
98 void swapDigits()
99 {
100 }
```

```
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day17.c Untitled2 x

101 int n, first, last, temp, p = 1;
102
103 printf("Enter number: ");
104 scanf("%d", &n);
105
106 last = n % 10;
107 temp = n;
108
109 while(temp >= 10)
110 {
111     temp = temp / 10;
112     p = p * 10;
113 }
114
115 first = temp;
116
117 n = n - first * p - last;
118     + last * p + first;
119
120 printf("After swapping = %d", n);
121 }
122
123 void palindrome()
124 {
125     int n, temp, rev = 0;
126
127     printf("Enter number: ");
128     scanf("%d", &n);
129
130     temp = n;
131
132     while(temp > 0)
133     {
134         rev = rev * 10 + temp % 10;
135         temp = temp / 10;
136     }
137
138     if(n == rev)
139         printf("Palindrome Number");
140     else
141         printf("Not a Palindrome Number");
142 }
143
144
145
```

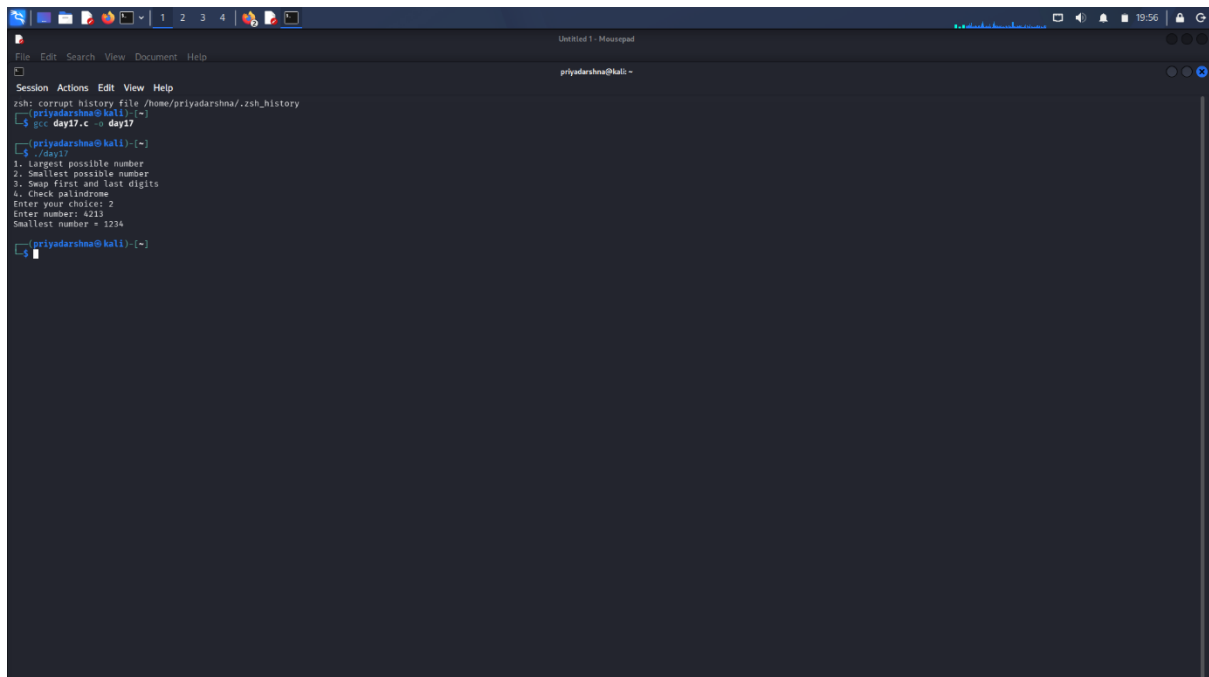
>Test case 1:



The screenshot shows a terminal window titled "priyadarshna@kali: ~" with a dark background. The terminal output is as follows:

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali]~$ gcc day17.c -o day17
[priyadarshna@kali]~$ ./day17
1. Largest possible number
2. Smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 1
Enter number: 4213
Largest number = 4321
[priyadarshna@kali]~$
```

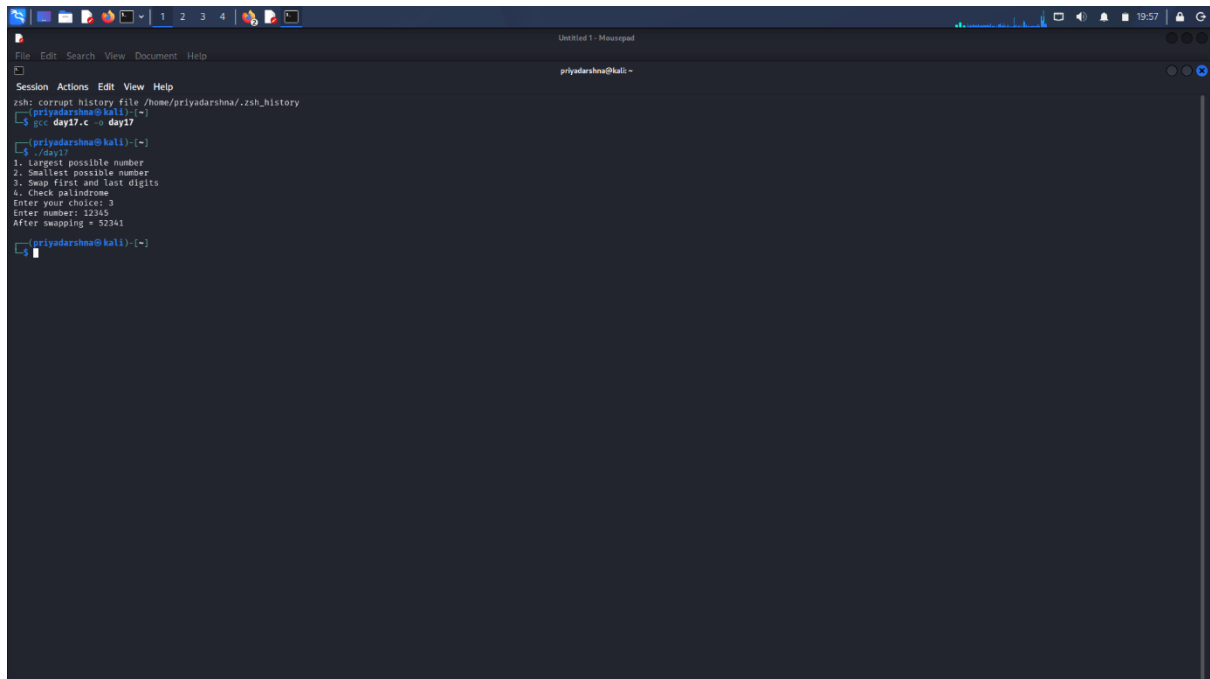
>Test case 2:



The screenshot shows a terminal window titled "priyadarshna@kali: ~" with a dark background. The terminal output is as follows:

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali]~$ gcc day17.c -o day17
[priyadarshna@kali]~$ ./day17
1. Largest possible number
2. Smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 2
Enter number: 4213
Smallest number = 1234
[priyadarshna@kali]~$
```

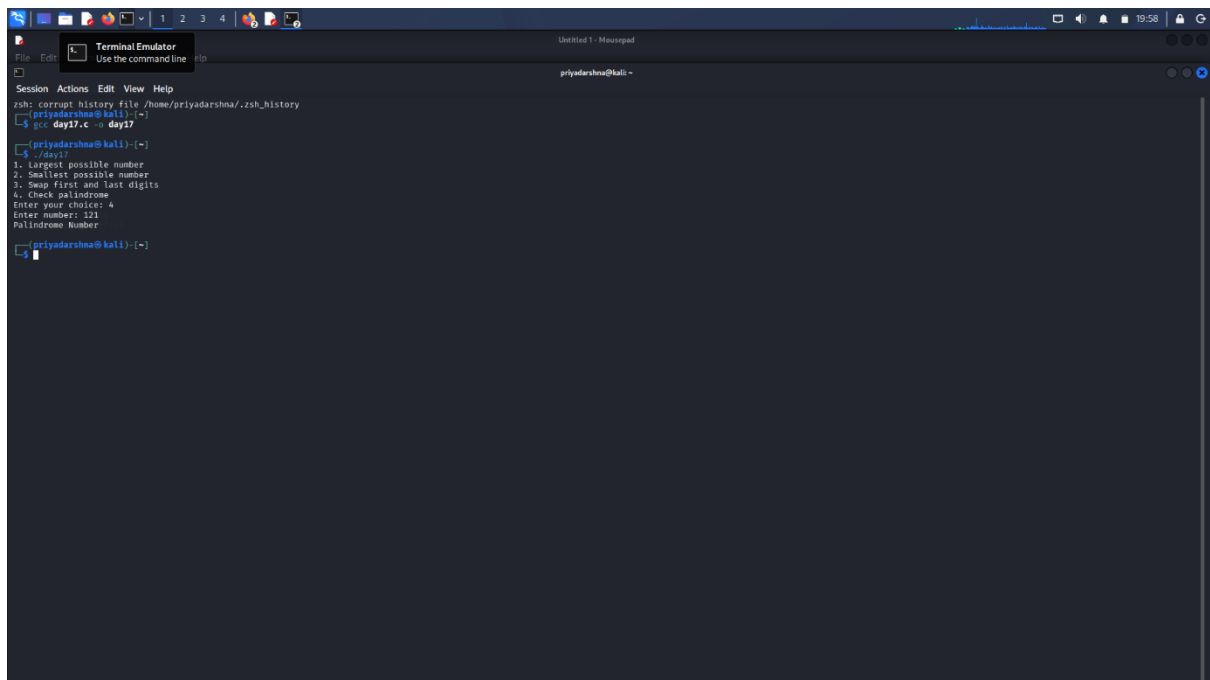
>Test case 3:



The screenshot shows a terminal window with a dark background. The prompt is `priyadarshna@kali: ~`. The user enters `gcc day17.c -o day17` and runs the program with `./day17`. The program displays a menu with four options: 1. Largest possible number, 2. Smallest possible number, 3. Swap first and last digits, and 4. Check palindrome. The user selects option 3. They then enter the number 12345. The program outputs "After swapping = 52341".

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
priyadarshna@kali: ~
$ gcc day17.c -o day17
priyadarshna@kali: ~
$ ./day17
1. Largest possible number
2. Smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 3
Enter number: 12345
After swapping = 52341
priyadarshna@kali: ~
$
```

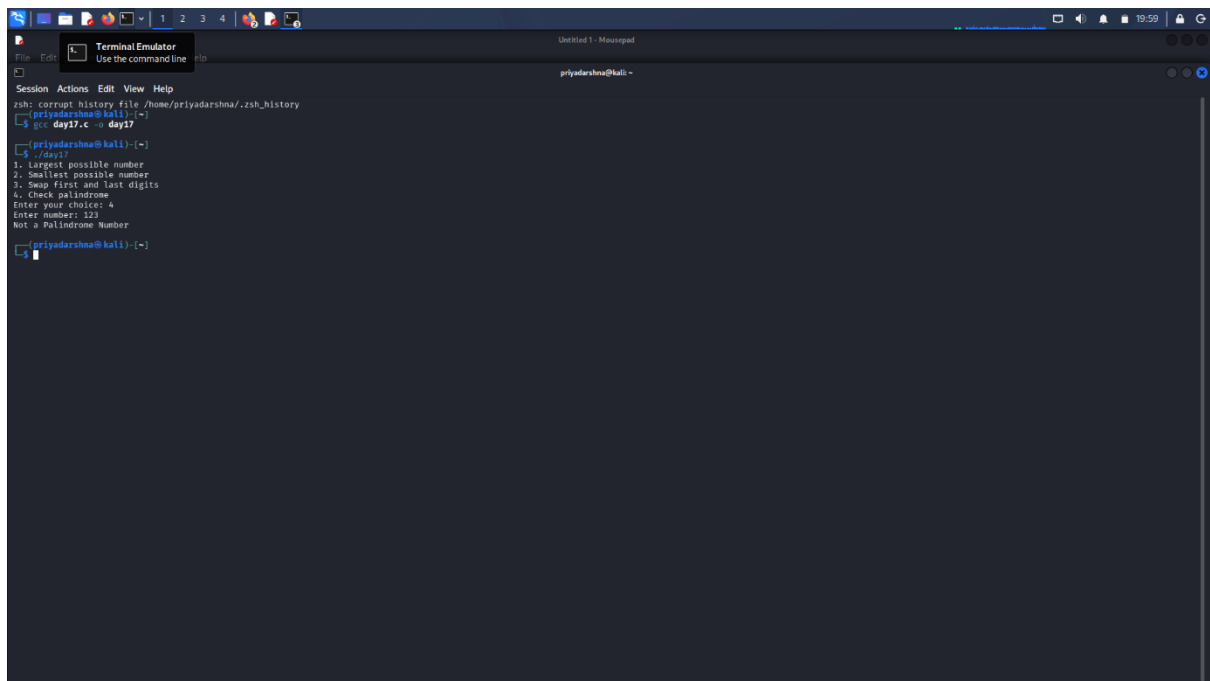
>Test case 4:



The screenshot shows a terminal window with a dark background. The prompt is `priyadarshna@kali: ~`. The user enters `gcc day17.c -o day17` and runs the program with `./day17`. The program displays the same menu as in Test case 3. The user selects option 4. They then enter the number 121. The program outputs "Palindrome Number".

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
priyadarshna@kali: ~
$ gcc day17.c -o day17
priyadarshna@kali: ~
$ ./day17
1. Largest possible number
2. Smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 4
Enter number: 121
Palindrome Number
priyadarshna@kali: ~
$
```

>Test case 5:



```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali]~$
$ gcc day17.c -o day17
[priyadarshna@kali]~$
$ ./day17
1. Largest possible number
2. Smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 4
Enter number: 123
Not a Palindrome Number
[priyadarshna@kali]~$
```